

AI assisted feedback system for transmit parameter optimization in Cardiac Ultrasound

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Abstract— Ultrasound image acquisition is often not repeatable with dependency on the skill of the operator to both acquire and optimize ultrasound images. Beyond arriving at the right scan planes, the user has to optimize a series of acquisition parameters like the frequency, depth, gain, TGC etc to obtain the optimal image. While varying acquisition parameters leads to varying image quality and clinical acceptability, image quality in ultrasound is very subjective and thus the choice of optimal image is often limited to the preference of the user. In this paper, we present an AI assisted feedback system to optimize ultrasound acquisition parameters. We also present a generic methodology for ‘Ultrasound Image Quality’ computation mimicking experts’ way of reading u/s images. We demonstrate this system to optimize two transmit parameters, namely frequency and depth on 4 chamber cardiac Ultrasound view images. We show that in >80% of unseen cardiac acquisitions, the proposed system is able to automatically select the optimal parameters, maximizing image quality.

Keywords— *Ultrasound Image Quality, Ultrasound Transmit parameters, Auto Frequency, Retina Net, Convolutional Neural networks*

I. INTRODUCTION

Ultrasound image acquisition depends heavily on skill of the operator. Desired image quality is dependent on multiple factors like right scan plan, subject variability or difficulty - resulting in haze/rib artefacts and usage of non-optimal acquisition parameters. The selection of optimal parameters involves sequential & iterative manipulations of various parameters like frequency, depth, gain, TGC etc. An automated solution for parameter selection can help the novice user in getting to right settings while also saving time for an expert user. However, setting the right transmit parameter for image acquisition setting is not a well-studied problem unlike the problem of scan plane detection, which has standard definitions of right plane. Fig 1 shows images with varying image quality across patient population. Optimal parameter selection is straightforward if ultrasound image quality can be estimated reliably. One of the fundamental challenges is that image quality in ultrasound imaging is not standardized and there is no metric or method exists today that model the multiple facets of ultrasound IQ. The other challenge is that each transmit

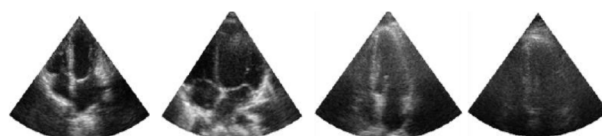


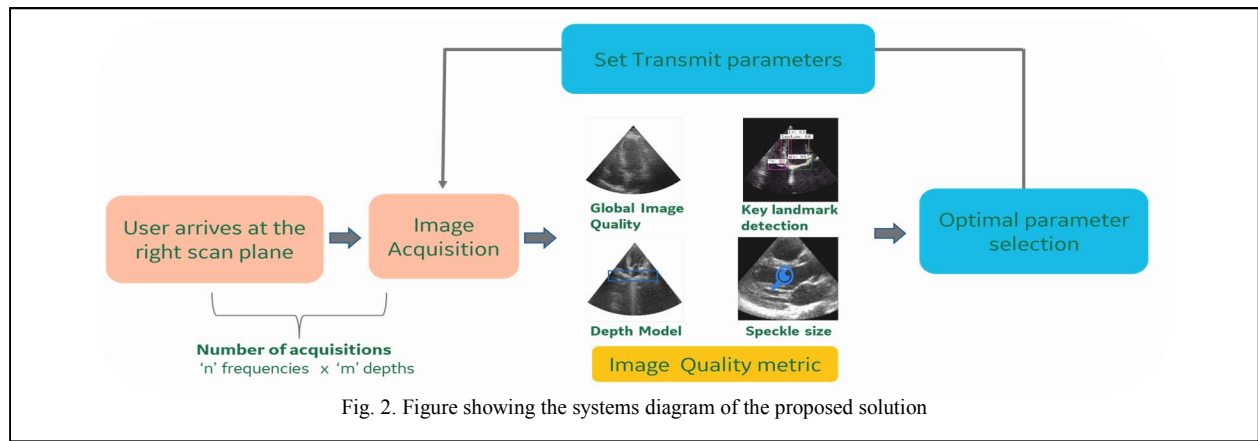
Fig. 1. The first 2 images are representative of good quality images and the next 2 images are of poor quality

parameter will impact the image quality in a different manner and their effects on resultant image quality is often confounding. In this paper, we propose a novel and state of the art methodology to address the problem of optimal parameter selection using deep learning.

Understanding the literature related to our paper can be broken into two areas of research 1) Image quality computation in ultrasound images 2) Auto-parameter optimization. As explained in an exhaustive survey in [1], measuring image quality in ultrasound involves quantifying a complex mixture of multiple features including presence of structures, overall noise in the image, absence of artefacts, etc. Listing all the works related to all these features is out of scope for this paper. In [2] and [3], authors propose computer vision-based approaches for object detection and delineation. In [4], authors provide a system for measuring quality in fetal head scans using texture-based segmentation. Most of the methods reported in patent literature [6-10] on parameter optimization are conceptual. A lot of methods target single parameter optimization – for eg. targeted to optimize for image gain [10] or TGC [9] as a post processing step on acquired dicom data. However, in [5] authors demonstrate a complete system of image quality computation and parameter optimization using manifold learning approaches. This work is the closest to the proposed method in terms of approach, however with key differences explained later. We provide an overview of the system in Section II and describe sub-components in later sections.

II. AUTO-PARAMETER SELECTION – SYSTEMS VIEW

In this paper, we present a system to optimize ultrasound transmit parameters by an AI assisted feedback system. The proposed system shown in Fig 2 consists of :



- 1) An acquisition system that is designed to acquire ultrasound images from combinations of multi-parameter settings in near real time.
- 2) A forward loop that computes Ultrasound Image Quality on the acquired multi-parameter data.
- 3) A feedback system that selects the transmit parameters generating optimal image.

A. Image Acquisition Setup

The data acquisition protocol involves triggering several transmit events with all possible combinations of parameters automatically and acquiring the corresponding cine-loops. To begin the acquisition process, the user arrives at the right view (ex. 4CH view) and the system triggers data acquisition as per a predefined scan table. As per our data collection protocol, images from 4 frequencies (1.4, 1.7, 2.0, 2.03 MHz) and 3 depths (10cm, 16cm, 30cm) are acquired from each patient.

B. Forward loop: Image Quality metric

We propose to model different properties of image quality through a multitude of neural networks. The composite image quality metric is arrived as the ensemble of outputs of these neural networks. In this paper, we demonstrate neural networks aimed to measure 1) Image noise & contrast 2) Image content 3) Speckle characteristics.

C. Feedback loop: Optimal parameter selection

The final component of the system is a feedback loop, that selects optimal acquisition parameters from the computed image quality followed by a selection logic. Once the optimal transmit parameters are selected, the algorithm automatically triggers the transmit system with these optimal values.

II. METHODS

Our image quality estimation is inspired by how a cardiologist reads ultrasound images and is designed to model important characteristics that they look for. A cine loop is deemed to be of good quality if –

- 1) image noise and artifacts are minimal
- 2) all structures well seen
- 3) meets the definitions of right scan plane

We explain the effect of varying depth and acquisition frequency and how it affects image quality.

• Effect of varying Depth on Image Quality

Optimal transmit depth is the right depth at which the farthest point in the scan sufficiently fits all the key landmarks required for the view. For ex. Fig 4 shows images acquired at varying depths and 17cm depth is the optimal setting. Wrong depth settings (Fig 3) can render the image unusable for specific applications that require accurate quantification, despite the

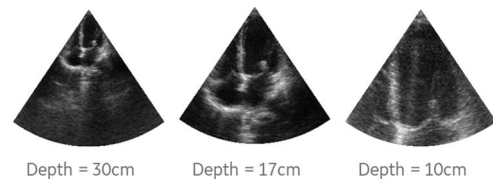


Fig. 3. Images showing 4CH cardiac view acquired at varying depth

presence of key structures.

• Effect of varying frequency on Image Quality

As the frequency of acquisition is increased from low to high, due to ultrasound beam attenuation, structures in the far field and thinner structures like valves start to diminish. In addition, the speckles get finer at highest the frequency, whereas the contrast is the highest at lower frequencies. So, a tradeoff between speckle size and attenuation is a key factor to arrive at optimal frequency. So, our solution hypothesis is – “The optimal transmit frequency is the highest frequency at which all the key landmarks and segments clearly seen”.

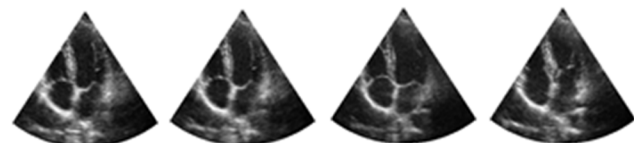


Fig. 4. Effect of changing frequency on the Image Quality with four different values – 1.4, 1.7, 2.0, 2.3 MHz (left to right)

Fig 4 highlight these trade-offs with varying frequency.

As discussed earlier in Section I, apart from acquisition parameters, several factors contribute to the image quality

numbers rated by an expert cardiologist. It is important to separate these two effects i.e. the population specific variability and patient specific variability. To this effect, we have developed a composite system of modules modeling different factors. The first module is a global Image Quality indicator tend to capture the population specific IQ and the next two modules are key landmark detector and a depth model modeling patient specific IQ.

A. Global Image Quality indicator

This is a population-based IQ indicator, that is learned to classify a given cine loop on a scale of 1-3 with (1 – bad, 2 – moderate, 3 – good). A multi-channel CNN (Fig 5) is designed to correlate cardiac loops (containing one cardiac cycle) to expert annotation on overall image quality. This is a global

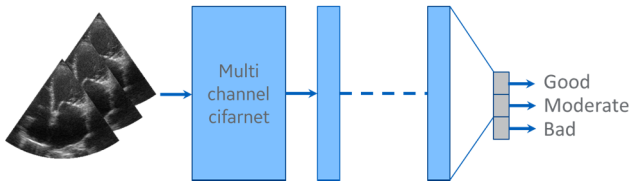


Fig. 5. Multichannel CNN architecture with cifar backbone to rate cine loop quality

image quality metric that rates the cine loop independent of the source of variability (ex. frequency, depth, scan plane). This metric captures the image contrast, image content and image noise but doesn't explicitly measure source of deterioration of image quality.

B. Key Landmarks detector

In standard cardiac views (4ch, 2ch, SAX, PLAX or 3ch), there are key structures that need to be present with enough contrast as the frequency is varied and our framework of landmark detection is very generic and can be extended to all standard cardiac views.

In 4CH view, key structures of interest are 1) Mitral valve 2) Tricuspid valve 3) Lateral wall 4) Septum 5) clear delineation of chambers. The objective of the key landmark detection module is to provide a metric which quantifies the image based on the presence of key structures. Since this task can be mapped to an object detection problem, a neural network-based model is used for solving it. We experimented with two different object detection networks – Faster R-CNN and RetinaNet.

RetinaNet [12] architecture is a popular object detection network that attacks the extreme foreground-background class imbalance problem during training. The architecture of this model (Fig 6) is composed of a backbone feature pyramid convolutional network, followed by two subnetworks – a classification subnet and a bounding box regression subnet. Faster R-CNN [11] belongs to two-pass object detector family whose detection accuracies are superior while suffering in speed. We evaluated both Faster R-CNN and RetinaNet architectures for detecting four important landmarks – mitral valve, tricuspid valve, septum and lateral wall from cardiac 4Ch view. We chose RetinaNet architecture for its speed and comparative detection performance. We explain details of RetinaNet training and performance in next section.

C. Depth Model

The depth model uses the Key landmark detection module mentioned above with a modified post processing step to select the optimal depth. A 4CH view image acquired at optimal depth, is expected to have the valves located approximately at half the acquisition depth. A spatial prior on the detected landmarks is used to compute IQ_{depth} .

D. Composite Image quality and Optimal parameters selection

The overall Image Quality metric is defined as a weighted combination of Global IQ and Structure IQ –

$$\text{Combined IQ} = a * IQ_{globalIQ} + b * IQ_{depth} + c * IQ_{structure} \quad (1)$$

We compute Combined IQ on all acquisitions coming from combinations of parameters and select the optimal setting from the acquisition with maximum combined IQ. The combination factors a,b,c were set heuristically following experiments.

III. EXPERIMENTS AND RESULTS

A. Data acquisition and curation

The data used in this paper were collected using GE VividE95 Cardiovascular Ultrasound scanner and M5ScD sector probe. From a total of ~320 subjects, a trained sonographer acquired 4ch cardiac cine loops 4 frequencies and 3 depths. An expert cardiologist reviewed each of the 12 loops per subject rating them on a scale of 1-5 and also picked the optimal image based on factors discussed before.

B. Global Image Quality indicator

A multi-channel CNN model with CIFAR backbone is used for determining the Global IQ. The model is trained on 530 cine loops, with 4 images of 256x256 resolution from each loop. This is a 3-class classification network which rates the image quality as good, moderate or bad. This network is tested on 75 unseen cine loops and achieves 80% accuracy as shown in Fig 7.

C. Key Landmarks detector

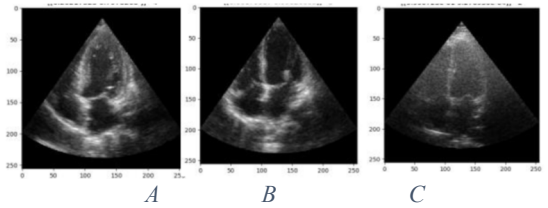


Fig. 7. Example images on which Predicted Global IQ rating from multi-channel CNN model and the corresponding ground truth rating matched. A) and B) are good class images C) – Bad class image

We compared two architectures for landmark detection 1) RetinaNet – single-shot detector 2) Faster RCNN – two-pass detector. For our experiments, we found that faster RCNN had lower inferencing speed of 2-3 frames per second with comparable detection accuracy of ~80% on test data. Since,

practical time considerations are important factors of successful adoption of this approach, we proceeded with RetinaNet and details of our training procedure is described next.

An open source implementation of RetinaNet based on PyTorch [13] was used. The model was trained on 2000 images annotated with the bounding box co-ordinates of the four landmarks. The model was tested with different backbones but ResNet50 followed by a Feature Pyramid Network (ResNet50FPN) which is pre-trained on ImageNet1k dataset worked well for our dataset. This network was trained for 10,000 iterations with a batch size of 8, 9 anchors at each spatial location corresponding to 3 different scales and aspect ratios respectively. Confidence score aggregation and non-maximal suppression were used to identify final detections by pooling over all relevant detections for every object. Parallel processing on GPU using CUDA API and conversion of trained model from Pytorch to TensorRT using ONNX intermediate representation yielded highest inferencing speeds upto 100 frames per sec on P2000 graphics card - a significant speed-up over Faster RCNN.

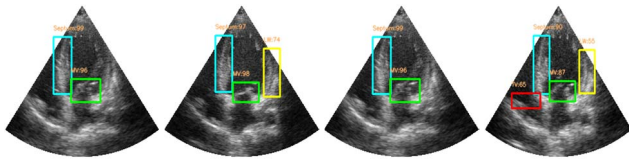


Fig. 8. Example images showing detected landmark using RetinaNet model

Fig 8. shows examples of key landmark detection results. As RetinaNet is less sensitive to weak/faint structures, it proved suitable for our problem. Ideally, the image quality score based on landmark detection should be high only if the landmarks are distinctly visible, which was satisfied naturally by RetinaNet. We needed to set a minimum threshold on confidence scores to consider them as successful detections. Because of lesser confidence scores generated by RetinaNet model, a threshold of 50% for an acceptable detection is set for our experiments.

D. Combined model results

The final composite image quality was arrived based on (1). After experimenting with few combination factors, we found that setting them equally was effective. Inferencing of the combined models was done on unseen data acquired from 26 patients at varying frequency and depth. Our proposed solution is able to accurately select the optimal parameters on 21 out of 26 patients matched with the expert's. In addition, the complete processing of the parameter selection takes less than 1s retiring any risks associated with practical implementation of such a solution on an ultrasound device. Fig 9 shows an example image where in the optimal depth is initially estimated followed on the lowest frequency setting, followed by selection of optimal frequency from cine loops acquired at optimal depth. Here in the example below, Optimal depth of 17cm and Optimal frequency of 1.7MHz is selected.

CONCLUSION

In this paper, we propose methodologies to address two major problems. 1) objective ultrasound image quality computation

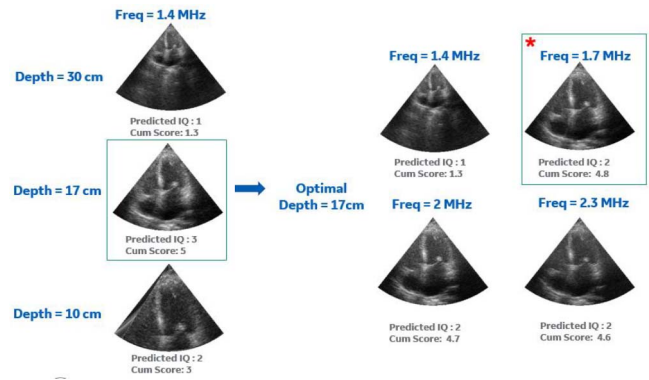


Fig. 9. Selection of optimal depth and frequency settings based on IQ score. The selected loop is also rated the highest by the expert

mimicking expert's intuition using multiple neural networks 2) automatic selection of optimal acquisition parameters for obtaining images with highest image quality. Standardization of ultrasound image quality and auto-parameter optimization are relevant problems which could help in standardizing ultrasound acquisitions and we feel our strong contributions will open more research in this area. In future, we plan to develop methods which model more characteristics like speckle properties and the effect of varying acquisition parameters modeling both - physics behind ultrasound acquisitions and utilizing expert's intuition in optimizing images.

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